Investigating Brain-LLM Alignment with Category-Deprived Text

Introduction Large Language Models (LLMs) over the years have achieved remarkable performance in several NLP tasks. Recent studies observe that LLMs that have been trained for next word prediction using large text corpora can be leveraged to significantly predict brain activity of subjects that listen to or read text (Jain & Huth, 2018, Toneva & Wehbe, 2019, Schrimpf et al., 2021, Oota et al., 2022). However, it remains unclear what linguistic features beyond sentence length, tree depth, top constituents, or tense, bigram shift can contribute to this effect (Jawahar et al., 2019, Conneau et al., 2018). Many experiments in cognitive neuroscience examine how particular language features are processed by manipulating the experimental stimulus and tracking the brain region BOLD response or the processing times (Pallier et al., 2011). Oota et al. (2022) has proposed a residual approach to study how eliminating linguistic information affects this alignment. However, we do not know of attempts of eliminating linguistic information at the source, from the training an LMM receives. Here we report preliminary results on precisely this approach.

Methodology We examine how removing different linguistic Part-Of-Speech (PoS) categories affects the alignment between LMs and brain responses. We targeted nouns, verbs, all content words and all function morphemes. To this end, we created multiple versions of a 12M English corpus sampled from Wikipedia, UKWAC and Gutemberg, removing token distinctions within a specific category (e.g. all the nouns were replaced with ”NNOUN”, all main verbs in “VVERBM”, etc.). We then trained from scratch two different language models (LSTM and GPT2-small) on the original corpus (LLM(**O**)), on the various category-corrupted corpora (LLM(**O-N**(ouns), LLM(**O-V**(erbs), LLM(**O-C**(content), LLM(**O-F**(unction)) and on other versions of **O** where we had randomly replaced words with a “WWORD” tag, in the same proportion as the various category-corrupted corpora (LLM(**O-R**(andom)**N**(noun), etc.); the last set was a baseline, to make sure that any variation in alignment was not simply due to the model having a “worse” training input). We then extracted language representations of spoken narrative stories that were used to collect fMRI responses of subjects while they were listening to these stories (Huth et al., 2016), fed the same text to our set of LMs, collected the representations from the last layer and calculate the alignment by learning an encoding model that maps the two representations (see Jain & Huth, 2018). Finally, we compared the alignment on the with and the linguistic information, evaluating on held-out test data.

Results The map of alignment deltas (O-x - O-Rx, x $\in$ {N, V, C, F}), see Fig. 1, shows a category-related shift. In particular: **(i)** O-N effects were stronger in left anterior temporal and parietal areas canonically associated with (lexico-)semantic processing (Binder et al. 2009).**(ii)** O-V effects were stronger in frontal and posterior temporal areas that are typically associated with verb processing (Lukic et al., 2021).**(iii)** O-C effects were somewhat stronger than those associated with the individual ablation of Ns and Vs, compatible with greater overall lack of lexico-semantic information, and partially reflecting the anatomical distribution of both N-less and N-less effects. **(iv)** O-F effects were the strongest and most widespread ones, possibly due to the vast morphosyntactic degradation, which resulted in ambiguous thematic interpretation.

These preliminary and qualitative results suggest that this simple approach holds promises for our understanding of the contribution of linguistic features in LM-to-brain alignment. Formal statistical analyses, including a broader set of PoS categories are planned for the near future.

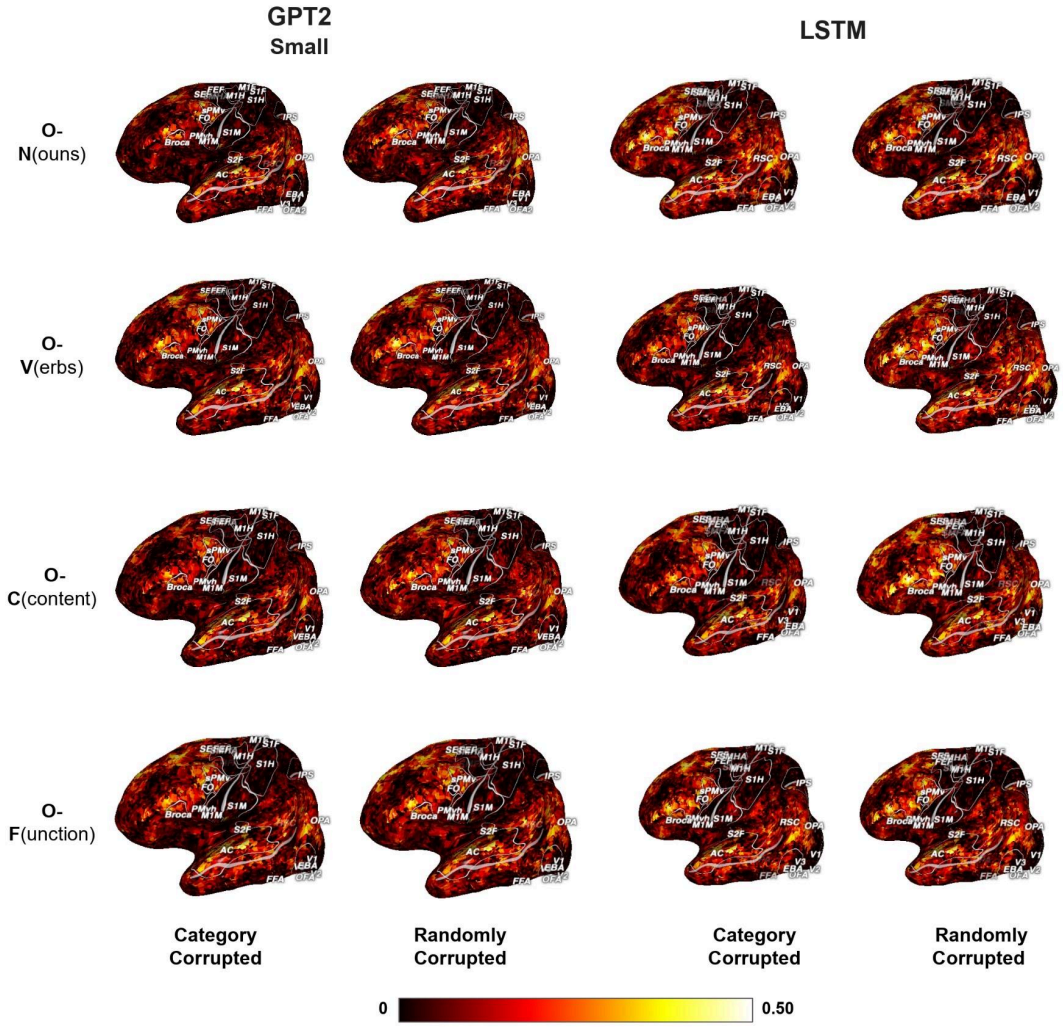

Figure 1: Brain alignment correlation maps Originally versus Randomly category-corrupted for (nouns, verbs, content words, and function words) with language models (GPT2-small and LSTM), revealing category-specific shifts in left anterior temporal/parietal (nouns), frontal/posterior temporal (verbs), and widespread (function words) regions.

Selected references

Binder, J. R., Desai, R. H., Graves, W. W., & Conant, L. L. (2009). Where is the semantic system? A critical review and meta-analysis of 120 functional neuroimaging studies. Cerebral Cortex, 19(12), 2767–2796. https://doi.org/10.1093/cercor/bhp055

Goldstein, A., Zada, Z., Buchnik, E., Schain, M., Price, A., Aubrey, B., Nastase, S. A., Feder, A., Emanuel, D., Cohen, A.,

Lukic, S., Borghesani, V., Weis, E., Welch, A., Bogley, R., Neuhaus, J., Deleon, J., Miller, Z. A., Kramer, J. H., Miller, B. L., Dronkers, N. F., & Gorno-Tempini, M. L. (2021). Dissociating nouns and verbs in temporal and perisylvian networks: Evidence from neurodegenerative diseases. Cortex, 142, 47–61. https://doi.org/10.1016/j.cortex.2021.05.006

Merlin, G., & Toneva, M. (2024). Language models and brains align due to more than next-word prediction and word-level information. Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, 18431–18454. https://doi.org/10.18653/v1/2024.emnlp-main.1024

Oota, S. R., Gupta, M., & Toneva, M. (2022). Joint processing of linguistic properties in brains and language models. arXiv. https://doi.org/10.48550/ARXIV.2212.08094